

CS3001 – Computer Networks

Quiz – 2A

Name	
Roll. No.	
Section	

Part 1 (8 Marks)

Suppose an institutional network has the following parameters:

- **Access link rate = 2 Mbps**
- **Average object size = 200 Kbits**
- **Average request rate from all browsers = 10 requests/sec**
- **RTT to the origin server = 1.5 sec and LAN delay is low (10ms).**

With local web cache: Cache hit rate = **50%** (all other requests go to the origin server). Ignore queueing when utilization ≤ 0.8 but assume **queueing delays infinite when utilization > 0.9 .**

	Without Cache	With Cache
Average access link utilization		
Average request delay		

Delay = Internet delay (RTT) + Access link transmission delay.

Part 2 (2 Marks)

You are tasked with designing a new application-layer protocol for a **real-time multiplayer VR game**. This protocol must support:

- Low-delay communication (on the order of tens of milliseconds).
 - High bandwidth fluctuations (due to both video streams and frequent positional tracking updates).
 - Authentication and encryption for all sessions.
 - No strict reliability guarantees for continuous position/video updates, but some **game-critical control data (e.g., match events, inventory changes)** must be reliably delivered.
1. Which transport protocol (**TCP, UDP, UDP with TLS, TCP with TLS**, or a **combination (hybrid)** of protocols) would you use and why?
 2. How should the state be maintained between client and server sessions (cookies vs server-managed state vs stateless) and why?

	Protocol(s)	Why?
1.		
2.		

Part 1 Solution:

Solution:

Step 1: Calculate traffic intensity (no cache).

- Arrivals/sec = 10 requests/sec
- Size/object = 200 Kbits
- Data rate required = $10 \times 200 \text{ Kbps} = 2000 \text{ Kbps} = 2 \text{ Mbps}$

Access link capacity = 2 Mbps $\rightarrow$ Utilization = $U = 2/2 = 1.0$

This is **100% utilization** $\rightarrow$ **unstable (huge delays due to queueing)**.

Step 2: With cache hit rate = 0.5.

- Only 50% requests go to the origin server.
- Effective arrival rate to Internet = $0.5 \times 10 = 5 \text{ req/sec}$.
- Data rate = $5 \times 200 \text{ Kbps} = 1000 \text{ Kbps} = 1 \text{ Mbps}$

Utilization = $U = \frac{1}{2} = 0.5$

This is safe (no queueing buildup).

Step 3: Expected delay (formula applied).

- Without cache:
 - RTT = 1.5s, transmission delay = $(200 \text{ Kbits} / 2 \text{ Mbps}) = 0.1 \text{ s}$
 - But utilization = 1 $\rightarrow$ **unbounded queueing delay** $\rightarrow$ **unacceptable**.
- With cache:
 - RTT = 1.5s, transmission delay = 0.1s
 - For 50% cache hits $\rightarrow$ delay only msec (local cache)
 - Average delay = $0.5 \times 1.6 \text{ s} + 0.5 \times 0.01 \text{ s} \approx 0.8 \text{ s}$

Final Answer:

- Without cache: Unstable (infinite delays).
- With cache: Average delay $\approx$ **0.8 sec**, acceptable.
 $\rightarrow$ **Cache drastically improves performance.**

Part 2 Solution:

Solution

1. Transport protocol:

- **TCP:** Unsuitable for real-time VR updates due to latency from retransmissions and head-of-line blocking.
- **UDP:** Good for low-delay, loss-tolerant data (position, video).
- **UDP+TLS:** UDP with encryption and authentication.
- **Hybrid:** Best choice. Use **UDP+TLS** for continuous, real-time game state (positions, video) where low latency is paramount and some loss is acceptable.
Use **TCP+TLS** for game-critical control data (e.g., match events, inventory) that requires guaranteed reliability.

Answer: Hybrid approach: **UDP+TLS** for real-time data, **TCP+TLS** for critical reliable data.

2. State maintenance:

- Stateless: Impractical for persistent player data.
- Cookies: Inadequate for complex game state and security.
- **Server-managed:** Best choice. The server maintains authoritative state (player position, inventory, match status) for each client, ensuring persistence and consistency.

Answer: Server-managed session state.